

TAILAI, China

WMO: 508440

Lat: 46.4012N

Lon: 123.4530E

Elev: 139

StdP: 99.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -25.3	(d) -32.5	(e) 0.2	(f) -25.8	(g) -30.3	(h) 0.2	(i) -23.2	(j) 8.5	(k) -11.1	(l) 7.4	(m) -10.9	(n) 1.6	(o) 290	(p) 0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.4	(c) 32.7	(d) 20.6	(e) 31.0	(f) 20.5	(g) 29.5	(h) 20.2	(i) 24.4	(j) 28.3	(k) 23.4	(l) 27.5	(m) 22.4	(n) 26.6	(o) 4.1	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.3	26.6	22.1	17.1	25.8	21.1	16.0	25.0	75.0	28.2	70.9	27.3	67.2	26.7	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	7.9	6.9	-29.2	36.0	4.0	2.0	-32.0	37.4	-34.3	38.6	-36.5	39.7	-39.4	41.1
			-29.4	25.6	3.8	1.2	-32.1	26.4	-34.3	27.1	-36.4	27.8	-39.2	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-16.2	-11.2	-2.5	7.7	15.6	21.7	24.0	22.1	15.9	6.6	-5.1	-14.3	(6)
(7)		DBStd	14.89	4.48	5.24	5.71	4.87	4.22	3.54	2.56	2.68	3.90	4.81	6.01	4.72	(7)
(8)		HDD10.0	3235	812	592	389	101	4	0	0	3	126	453	754		(8)
(9)		HDD18.3	5153	1071	826	647	320	107	9	1	3	91	363	703	1012	(9)
(10)		CDD10.0	1572	0	0	1	32	179	350	434	374	181	21	0	0	(10)
(11)		CDD18.3	449	0	0	0	1	23	110	177	119	19	0	0	0	(11)
(12)		CDH23.3	3828	0	0	0	19	268	1019	1479	877	163	4	0	0	(12)
(13)		CDH26.7	1218	0	0	0	4	89	381	491	224	28	1	0	0	(13)
(14)	Wind	WSAvg	3.2	2.6	2.9	3.7	4.2	4.1	3.3	2.9	2.8	3.0	3.2	3.0	2.6	(14)
(15)	Precipitation	PrecAvg	391	1	2	5	13	25	71	130	83	41	15	4	2	(15)
(16)		PrecMax	672	4	7	20	85	79	170	352	201	83	63	15	8	(16)
(17)		PrecMin	227	0	0	0	0	0	1	32	12	3	0	0	0	(17)
(18)		PrecStd	100	1	2	5	17	19	44	67	51	23	16	4	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	6.4	16.8	25.9	32.7	36.2	35.1	32.5	29.6	23.2	12.8	0.9	(19)
MCWB			-5.5	0.2	6.6	11.8	16.0	19.7	22.4	21.9	19.0	12.9	5.4	-2.4	(20)	
2%		DB	-5.4	2.6	12.3	22.2	29.3	32.9	32.6	30.5	27.0	19.6	9.0	-2.8	(21)	
		MCWB	-7.9	-2.2	4.1	10.0	15.1	19.4	22.1	21.6	17.6	11.2	3.1	-5.3	(22)	
5%		DB	-7.6	-0.3	9.3	19.6	26.3	30.7	30.9	29.1	25.2	16.6	5.8	-5.2	(23)	
		MCWB	-9.6	-3.9	2.4	9.0	14.2	18.9	22.0	21.2	16.6	9.4	1.3	-7.3	(24)	
10%		DB	-9.6	-2.7	6.7	16.8	23.8	28.8	29.3	27.7	23.3	14.4	3.0	-7.3	(25)	
		MCWB	-11.2	-5.7	0.7	7.5	13.0	18.4	21.7	20.6	15.5	8.1	-0.7	-9.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.5	0.3	6.8	13.0	18.2	22.8	26.1	25.1	21.7	14.6	5.6	-2.0	(27)
MCDWB			-2.3	6.0	15.7	23.0	27.2	29.6	29.5	28.8	26.1	20.1	12.0	1.0	(28)	
2%		WB	-7.8	-2.0	4.4	11.2	16.6	21.6	24.9	24.1	19.7	12.5	3.5	-5.4	(29)	
		MCDWB	-5.6	2.4	11.6	20.0	24.8	27.9	28.9	27.8	24.1	17.6	8.5	-2.8	(30)	
5%		WB	-9.5	-3.8	2.7	9.5	15.3	20.7	24.0	23.0	18.2	10.6	1.6	-7.3	(31)	
		MCDWB	-7.6	-0.3	8.7	18.6	23.5	26.9	28.0	27.0	22.8	15.4	5.6	-5.4	(32)	
10%		WB	-11.1	-5.6	0.9	8.0	14.2	20.0	23.1	22.0	16.8	8.7	-0.4	-9.0	(33)	
		MCDWB	-9.6	-2.7	6.2	15.8	22.6	26.0	27.1	25.9	21.6	13.7	2.7	-7.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	11.4	12.0	11.3	9.8	8.4	9.1	11.1	10.6	9.5	9.6	(35)
MCDBR			12.0	12.9	14.5	15.5	15.2	13.0	10.8	10.9	13.1	13.8	11.9	11.1	(36)	
MCWBR			10.0	9.5	8.8	7.9	6.0	4.3	4.1	4.4	5.8	7.2	7.7	8.9	(37)	
5% WB		MCDWB	11.9	12.8	14.0	14.0	12.2	9.8	8.3	8.5	10.6	11.9	11.6	10.9	(38)	
		MCWBR	9.9	9.6	8.6	7.5	5.8	4.0	4.2	4.3	5.5	7.3	7.6	8.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.312	0.377	0.466	0.538	0.590	0.543	0.483	0.424	0.390	0.328	0.300	(40)	
taud		2.212	2.187	2.073	1.896	1.786	1.772	1.933	2.067	2.139	2.109	2.192	2.176	(41)		
Ebn at Noon		786	848	834	787	743	706	736	766	778	737	726	720	(42)		
Edh at Noon		84	106	139	182	211	216	181	152	130	114	85	78	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.98	3.12	4.40	5.44	6.01	5.89	5.68	5.37	4.54	3.24	2.07	1.58	(44)	
(45)		RadStd	0.21	0.23	0.25	0.33	0.39	0.52	0.54	0.33	0.34	0.16	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	-2.56	-1.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page